

FinPilot

AI-Powered Investment Analysis Platform

Executive Summary

Prepared for: Grant & Investor Applications

March 2026 | v1.0 | Confidential

1. The Opportunity

Retail and institutional investors are overwhelmed by data. Despite access to thousands of financial data points, most investors lack the computational tools to systematically identify high-probability opportunities. FinPilot solves this by combining Deep Reinforcement Learning (DRL) with Large Language Models (LLM) into a single, automated investment intelligence platform.

Market Problem

- Over 58 million retail investors in the EU actively trade equities, yet fewer than 2% use algorithmic tools.
- Professional algo-trading systems (Bloomberg Terminal, FactSet) cost \$25,000–\$75,000/year — inaccessible to individual investors and small funds.
- Existing retail tools (TradingView, StockCharts) offer indicators but no AI-driven decision support or automated learning.
- The global algorithmic trading market is projected to reach \$31.4 billion by 2028 (CAGR 11.6%).

Our Solution

FinPilot is a cloud-native AI platform that automatically scans 1,148 NASDAQ symbols across 11 sectors, applies 19 trained DRL agents, synthesises signals via a multi-LLM router, and delivers actionable investment intelligence — without requiring any programming knowledge from the end user.

2. Technology & Traction

Platform Metrics

Sprint Development Cycles	22	DRL Models Trained	19
Stock Symbols Scanned	1,148	Market Sectors Covered	11
LLM Providers (Failover)	3	API Endpoints	11
Test Coverage	466 Tests	CI/CD Pipeline Jobs	7

Deep Reinforcement Learning Stack

FinPilot's proprietary DRL engine is built on Stable-Baselines3 with PPO/SAC/TD3/A2C algorithms trained in a Gymnasium-compatible market environment. Key technical achievements:

- Walk-forward training with 70/30 train/test split and 5-phase curriculum learning.
- Composite scoring formula: $(\text{Sharpe} \times 10) + (\text{Calmar} \times 0.3) - \text{overtrading penalty} - \text{negative return penalty}$.
- Optuna hyperparameter optimisation (30 trials per agent) with SQLite/PostgreSQL backend.
- PilotShield risk control: max position 75%, max leverage 1.5×, confidence threshold 65%.
- LearnableEnsembleWeights: EMA-based dynamic agent weighting combining regime prior and performance EMA.

Top-5 Model Performance (March 2026)

Model	Return	Max Drawdown	Sharpe	Composite Score
PPO Trend #1	+574%	46.5%	0.057	4.27
PPO Trend #2	+353%	37.7%	0.053	3.33
PPO Momentum #1	+300%	47.3%	0.063	2.23
PPO Trend #3	+270%	46.8%	0.043	2.16
PPO Momentum #2	+211%	41.1%	0.055	1.80

Note: Models are in active optimisation. Current dd_weight optimisation (0.3 → 0.8) target: MaxDD <20%, Sharpe >0.5 within 2 iterations.

3. Technology Stack

Layer	Technology
Frontend (Dashboard)	Streamlit v1.7.0 — 5-tab analysis interface
Frontend (Modern Web)	Next.js + TypeScript — Progressive migration
Backend API	FastAPI — 11 routers, JWT auth, SlowAPI rate limiting
AI/ML Engine	Stable-Baselines3 PPO/SAC/TD3/A2C + Gymnasium
LLM Router	Groq → Claude → Gemini (automatic failover)
Hyperparameter Optimisation	Optuna — 30 trials/agent, SQLite/PostgreSQL
Database	SQLite (dev) / PostgreSQL (production)
Broker Integration	Alpaca API — paper & live trading
Monitoring	Prometheus + Grafana + Sentry
Infrastructure	Docker Compose + GitHub Actions CI/CD (7 jobs)
Data Sources	yfinance + Polygon.io + News API + On-chain
Security	JWT HS256, bcrypt, detect-secrets, Bandit/Safety

4. Business Model

Revenue Streams

Tier	Description	Price	Target
Free	5 daily scans, basic signals	€0/month	User acquisition
Pro	Unlimited scans, LLM analysis, DRL signals	€29/month	Retail investors
Premium	API access, custom backtest, early model updates	€99/month	Active traders
API / B2B	Per-call usage for fintech developers and funds	€49/10K calls	Developers & SMEs
White-label	Custom deployment + model fine-tuning	€5K–20K	Boutique funds

Revenue Projections (3-Year)

Year 1 (2026)	€85,000	Year 2 (2027)	€340,000
Year 3 (2028)	€1,200,000	Break-even	Month 18

5. Funding Request & Use of Funds

Total Funding Sought: €500,000

Category	Amount	Purpose
Cloud Infrastructure & Deployment	€80,000	AWS/Azure migration, PostgreSQL, Redis, auto-scaling
AI/ML Research & Development	€150,000	DRL model optimisation, backtesting engine, real-time feeds
Product Development	€120,000	Next.js frontend, mobile app, API marketplace
Regulatory & Legal Compliance	€50,000	MiFID II compliance, Austrian FMA advisory, IP protection
Sales, Marketing & Growth	€70,000	Content marketing, B2B sales, partnership development
Operations & Team Expansion	€30,000	DevOps, customer support, administrative

Grant Programs Targeted

- Austria: FFG Basisprogramm 2026 (up to €3M, 70% funding rate) — AI research component
- Austria: aws Preseed / aws Gründungsfonds II (€72M+ fund, active investment phase)
- Austria: riz-up GENIUS Ideenpreis 2026 (deadline: 26 April 2026, Niederösterreich)
- EU: EIC Accelerator 2026 (up to €2.5M grant + €10M equity) — deep tech AI category

- USA: NSF SBIR Phase I (up to \$305,000 non-dilutive) — AI systems focus area

6. Competitive Advantages

Advantage	Details
Proprietary DRL Engine	19 trained models, Optuna HPO, walk-forward validation — 22 sprints of R&D
Multi-LLM Failover	Groq → Claude → Gemini router ensures 99.9% AI analysis availability
Full-Stack Production Build	7-job CI/CD pipeline, Docker, JWT auth, Prometheus monitoring — enterprise-grade from day one
Speed to Market	Streamlit dashboard deployable in days; Next.js migration underway for scalable UI
Proprietary Data Signals	1,148-symbol scanner + on-chain data + news sentiment — multi-source alpha generation
Regulatory Positioning	Architecture designed as 'analysis tool', not 'investment adviser' — MiFID II compliant framing

7. Team & Vision

FinPilot was developed by a founder with deep expertise in AI/ML engineering, financial markets, and full-stack software development — evidenced by 22 consecutive sprint cycles building an enterprise-grade platform from scratch. The project demonstrates a rare combination of academic rigour (DRL research, Optuna HPO, walk-forward validation) and engineering discipline (CI/CD, Docker, testing, security).

12-Month Vision

- Q2 2026: Cloud deployment (AWS/Azure), PostgreSQL migration, paid beta launch
- Q3 2026: EIC Accelerator application submitted; NSF SBIR Phase I filed
- Q3 2026: First 500 paying subscribers; 3 B2B API contracts signed
- Q4 2026: Real-time WebSocket data feeds live; mobile app beta
- Q1 2027: MaxDD <20% across all DRL models; Sharpe >0.5 achieved
- Q2 2027: White-label deployment for first institutional client

8. Contact

FinPilot

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This document is confidential and intended solely for grant evaluation and investor review purposes.